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Current Density Image Prediction (CDIP) from SQUID-on-tip Magnetic Field Data

Chico Demmenie (4451856)

Supervisors:

Dr. Daan M. Pelt & Dr. Kaveh Lahabi

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Leiden Institute of Advanced Computer Science (LIACS)

liacs.leidenuniv.nl

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Abstract

This thesis applies the UNet deep learning architecture proposed in a previous paper to SQUID-On-Tip (SOT) images to reconstruct the current density inside the underlying sample. It found that the more difficult task of current reconstruction from SOT, compared to Nitrogen Vacancy (NV) imaging, was still possible using the same architecture, that this outperformed previously popular methods such as Fourier Analysis, and that the performance can vary widely depending on how the training data is created.

Contents

1	Introduction	1
2	Background Concepts	3
2.1	SQUID-on-tip	3
2.2	The forward problem: simulating magnetic fields	3
2.3	Neural Networks, CNNs and UNets	5
2.4	Hyperparameters and Bayesian Optimisation	7
3	Related Work	9
3.1	Current Reconstruction by Fourier Analysis	9
3.2	Bayesian Inference Method	9
3.3	MAGIC UNET	9
3.3.1	Normalisation	10
3.3.2	Network Architecture	10
3.3.3	Training	10
4	Methods	11
4.1	Data	11
4.1.1	SQUID Angle conversion	12
4.1.2	SQUID Tools	12
4.1.3	Pre-processing	13
4.2	Model Architecture	13
4.3	Training	13
4.4	Hyperparameter Optimisation	13
4.5	Evaluation	14
4.5.1	RMSE	14
4.5.2	SSIM	14
5	Experiments & Results	15
5.1	Comparison between CDIP and MAGIC UNet	15
5.1.1	Quantitative results	15
5.1.2	Qualitative results	15
5.2	Comparison between CDIP and Fourier Analysis	16
5.2.1	Quantitative results	16
5.2.2	Qualitative results	18
5.3	Comparison between <code>squidtools</code> and angle conversion	19
5.3.1	Quantitative results	19
5.3.2	Qualitative results	19
6	Discussion	20
6.1	Contribution	22
6.2	Limitations	22
6.3	Future Work	22

7	Conclusion	24
8	Reflection	25
8.1	AI use	25
9	Acknowledgements	26
	References	30
A	Appendix	31
A.1	Technologies benefitting from CDR	31

1 Introduction

A major problem in quantum chip design and manufacturing [35], as well as in superconductors [10], semiconductors [4], and many other technologies (App. A.1), is the difficulty of identifying and understanding material and manufacturing defects. This inspection and diagnostics problem manifests primarily as an inability to see where the current flows through the material [7][46]. Often, the total current and its start and end points are known, but not where and how it flows between them.

Magnetic imaging techniques are a popular remedy for this problem because they are non-invasive [33]. Recent technological developments, such as Nitrogen-Vacancy (NV) centres in diamond[8], have allowed for higher-resolution magnetic imaging than previous techniques, such as magnetised atomic force microscopy tips [38] and field-sensitive electron microscopy [23], allowing a more accurate picture to be built of what goes on within a sample. Another such technique, SQUID-on-tip (SOT), stands out for its ability to probe magnetism, temperature, and geometry simultaneously, providing a comprehensive picture across all three domains [35]. This property makes it an ideal technique for understanding the transport of charge and heat in exotic materials, such as those dominated by quantum phenomena.

While SOT and NV imaging do a good job of measuring magnetism, the ultimate goal is to understand the current distribution within the material, thereby enabling a deeper comprehension of many-body quantum phenomena and facilitating easier diagnostics in quantum hardware design and superconductors [46]. Traditional methods for Current Density Reconstruction (CDR) from magnetic field images, such as the Fourier transform [37] or Bayesian inference [11], do a poor job of representing the level of detail required for delicate tasks as mentioned above [29], especially when used to resolve features on the same length scale as the sensor size and/or the sensor-sample distance. Improvements in a range of machine learning techniques enable more accurate image reconstruction, as demonstrated recently using a UNet architecture [36] on NV imaging data in the "MAGIC UNet" paper by Reed *et al.* (2025) [33].

This thesis aims to apply this architecture to SOT images. This can be achieved either by applying a simple angular transformation to existing MAGIC UNet data or by simulating a SQUID [34]. Another aim is to enhance current density reconstruction by applying Hyperparameter Optimisation (HPO) techniques [14].

The main questions this thesis will seek to answer include:

- Can the techniques used in the MAGIC UNet paper be adapted to SOT data?
- Does this setup improve over the previous Fourier and Bayesian Inference methods?
- Does this model perform better with data from a simulated SQUID than with a simple angular transformation of the previous NV data?

To answer these questions, the MAGIC UNet architecture was adapted to be able to predict current density, with x and y components, from only a single component image taken at an angle in the way that a SQUID would measure it. The data used was the same as in MAGIC UNet, which was generated using COMSOL Multiphysics and then transformed to mimic SQUID data in one of two ways, either by a simple mathematical transformation or by simulating a SQUID. The data is then preprocessed before being used to train the model. A few different Hyperparameter Optimisation (HPO) runs were tried, using Bayesian Optimisation (BO), before a standard set of

hyperparameters was settled on, and all training runs after that used the same set. The models were evaluated using two different metrics, Structural Similarity Index Measure (SSIM) and Root Mean Squared Error (RMSE), which means that a comparison between this method, MAGIC UNet and Fourier analysis is possible.

The experiments in this thesis included a comparison between CDIP and MAGIC UNet, which showed that both models perform comparably, with MAGIC UNet perhaps slightly outperforming CDIP, probably because the task is more difficult for CDIP. The second experiment was a comparison between CDIP and Fourier Analysis, which showed that CDIP clearly outperforms Fourier. Lastly, there was a comparison between CDIP models trained on two different types of converted data. This showed that a model trained on simulated SQUID data is very sensitive to the parameters used to simulate the data, while the model trained on simple angle converted data doesn't really have this problem.

This thesis showed that this kind of deep learning architecture applies to SOT data, and that it performs comparably on this kind of data to NV imaging data. It also showed that this architecture outperforms previous methods of current reconstruction and that the method of training data transformation can drastically affect the performance of the model.

2 Background Concepts

This thesis relies on several key concepts, such as SOT, which creates the real-world images, the Biot-Savart law, which can find the magnetic field around a defined current, neural networks in general and the UNet model specifically, which can predict current density from a magnetic field, and Bayesian Optimisation, which can optimise a neural network. These concepts will be explained here.

2.1 SQUID-on-tip

The real-world images used in this thesis were acquired with a tapping-mode Superconducting QUantum Interference Device (SQUID)-on-tip (SOT) microscope as detailed by Rog *et al.* (2026) [35]. This uses proximity Josephson junctions [26] and Atomic Force Microscopy (AFM) [28] to measure the magnetism, thermal dissipation, and topography of a sample. This SQUID is fabricated at the end of an AFM cantilever, with a fixed angle of 62° relative to the sample. This orientation allows the SQUID to sample 2 components of the magnetic field and provides an additional degree of freedom to tune the SQUID. The SQUIDS are produced on commercial AFM cantilevers strongly coupled to a quartz tuning fork, rather than on standard cantilevers, to avoid the need for an optical readout of the height above the sample, which would otherwise risk local heating and other interference.

Other key features of the nanoSQUID probes used here include SNS junctions, which are non-hysteretic at all temperatures [16], and multilayer heterostructures, allowing for flexible sensor customisation, including a choice of sensor size between 50nm and several microns, depending on the desired trade-off between field sensitivity and resolution. Overall, this design is more sensitive to in-plane magnetic signals than other SQUID setups, such as SQUID-on-lever, thereby enhancing the ability to resolve small currents [35].

2.2 The forward problem: simulating magnetic fields

The forward part of the physics problem, finding the magnetic flux density $\mathbf{B}$ at position r resulting from some current flowing along a closed/curved wire, is a straightforward problem which can be solved analytically using some calculus, as shown in Eq. 1:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_V \frac{(\mathbf{J}dV) \times \mathbf{r}'}{|\mathbf{r}'|^3}, \quad (1)$$

where $\mathbf{r}'$ is the vector from dV to point r , dV is the volume, $\mathbf{J}$ is the current density vector in the volume, and μ_0 is the magnetic constant [18].

This Biot-Savart law [47] is the standard starting point for generating synthetic data. Using known geometry, such as a network of wires or a physically interesting sample, and a known current distribution therein, the field around the sample can be calculated [37]. Because the prior current distribution is known, this can be used as a "label" or "ground truth", with the resulting magnetic field image being the "input" data, allowing a machine learning process to learn the relationship between the magnetic field and the underlying current distribution [33]. This can then be applied to field images collected from real samples, reconstructing current distributions which are unknown or uncertain.

The problem with using such data to train a model for SOT is that it is a poor representation of what the SQUID actually measures. The SQUID doesn't measure the magnetic field at an infinitesimal point; it probes the flux through an extended loop with a significant radius and often at a larger distance, meaning that images are less clear and much noisier than those created using Biot-Savart.

In effect, the measured signal $S(x, y)$ from a real SQUID is connected to the underlying current distribution $\mathbf{J}(x', y')$ through a convolution [34] as shown in Eq. 2, which is equivalent to Eq. 1, for a particular choice of $\mathbf{K}$:

$$S(x, y) = \int dx' dy' \mathbf{K}(x - x', y - y', \theta) \cdot \mathbf{J}(x', y'), \quad (2)$$

where (x, y) are the image coordinates, and (x', y') are the sample coordinates. Here, the convolution relies on a vector of kernels $\mathbf{K}$, which encode both the Biot-Savart law and the spatial size of the SQUID; this then depends on some hyperparameters θ , such as sensor size, orientation and height, as shown in Figure 1. So, to reconstruct the current density of a real image, the model must be trained on data that simulates a SQUID, rather than a direct approximation of the magnetic field [34].

Figure 1(a) shows the geometry of the SQUID above the current, which sits at $z = 0$. This current induces a magnetic field $B(r)$ at all positions $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$. The centre of the SQUID is at height $z = h$; it has inner and outer diameters of ρ_1 and ρ_2 , respectively, and is tilted about the y -axis by an angle ϕ . Therefore, the SQUID reads $B \cdot \hat{n}$, which contains both the x and z field components. The response of the SQUID is given by the convolution of the Biot-Savart field with a window function $W(\rho)$, which encodes the SQUID's sensitivity, as shown in Figure 1(b). If the SQUID has a uniformly distributed bias current and the screening can be neglected, then the window function is a constant at radii $\rho < \rho_1$ and linearly tapers to zero for $\rho_1 < \rho < \rho_2$ [34]. These approximations result in the SQUID response function in Eq. 3:

$$S(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint dx' dy' d\alpha d\rho W(\rho) \left(\mathbf{J}(x', y') \times \frac{\mathbf{p}}{p^3} \right) \cdot \hat{n}, \quad (3)$$

where (ρ, α) are polar coordinates in the SQUID plane and $\mathbf{p}$ is the displacement vector $\hat{\mathbf{x}}(x - x' + \rho \cos \alpha) + \hat{\mathbf{y}}(y - y' - \rho \sin \alpha \cos \phi) + \hat{\mathbf{z}}(h + \rho \sin \alpha \sin \phi)$.

This calculation is performed in a Python package called `squidtools` [34], which is used here to simulate the SQUID over existing current-density data, creating a more accurate representation of the image a SQUID would produce.

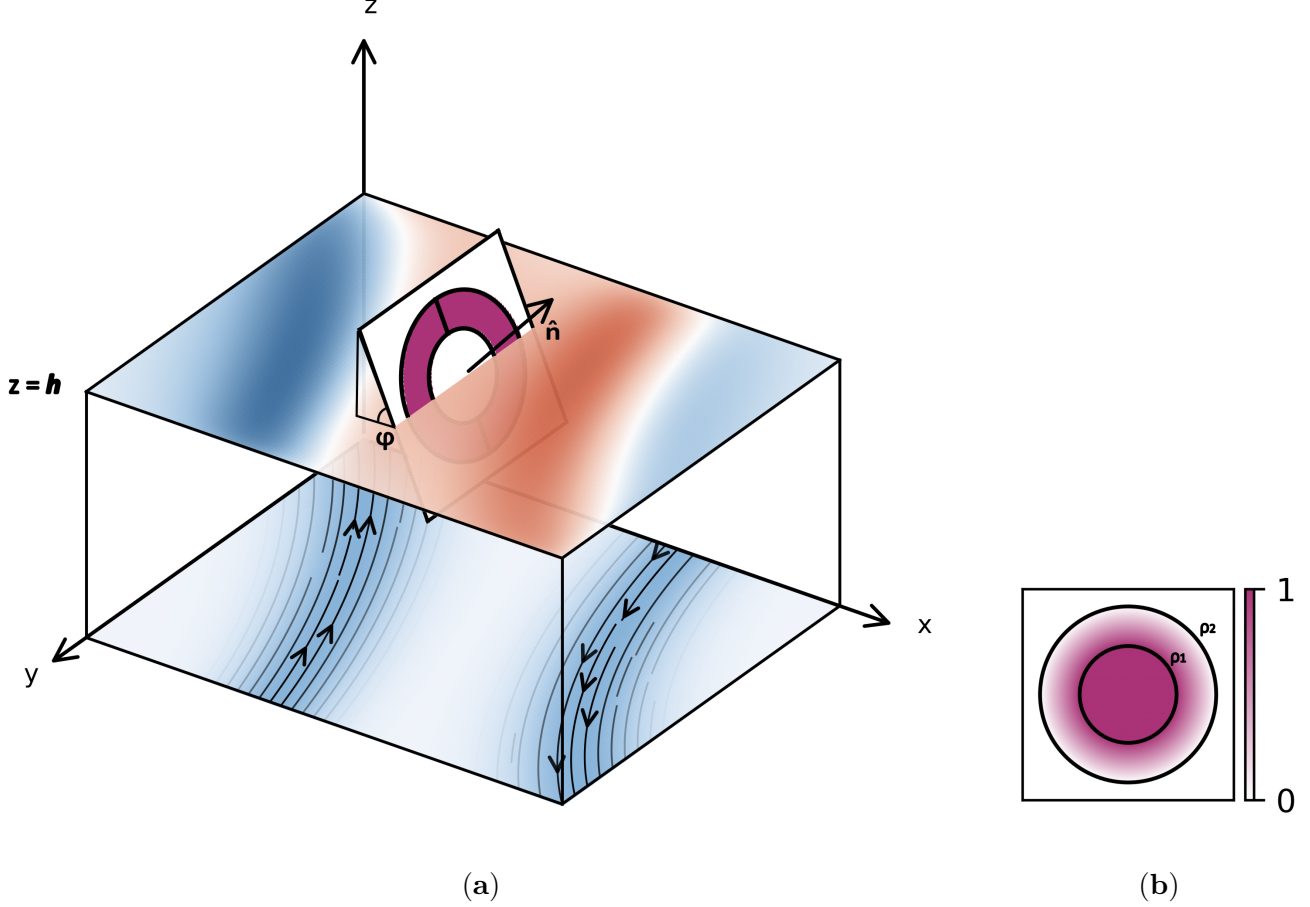


Figure 1: **Illustration of the SQUID geometry** courtesy of Rog (2026)[34] (a) Schematic overview of the SQUID and its position relative to the magnetic field and the current. The current at $z = 0$ generates a magnetic field $\mathbf{B}$ above. The SQUID at $z = h$ is tilted about the y -axis by ϕ and records the field component $\mathbf{B} \cdot \hat{\mathbf{n}}$. (b) The window function of the SQUID $W(u, v)$ weights the field through the SQUID. The function is unity for $\rho < \rho_1$ and linearly tapers to zero for $\rho_1 < \rho < \rho_2$.

2.3 Neural Networks, CNNs and UNets

A neural network is a machine learning model that mimics the basic functions of the human brain by connecting multiple interconnected nodes or neurons to approximate some complex function [40]. The simplest neural network is a so-called "perceptron" with input and output nodes corresponding to the input and output dimensions, plus one "hidden" node that mimics a neuron. Each of these steps also has weights and biases, which determine how much each node should contribute towards the output, as shown on the left of Figure 2. Perceptrons suffer from the problem that they are only able to approximate singular, often linear functions, but if many of these perceptrons are combined in multiple layers, then the resulting Multi-Layer Perceptron (MLP) can approximate more complex patterns, as seen on the right of Figure 2 [32]. MLPs are a type of Deep Learning model, meaning that there are multiple layers of hidden nodes stacked one after the other, as shown on the right of Figure 2. These "deep learning" networks often perform better than networks which

use fewer, wider layers [6].

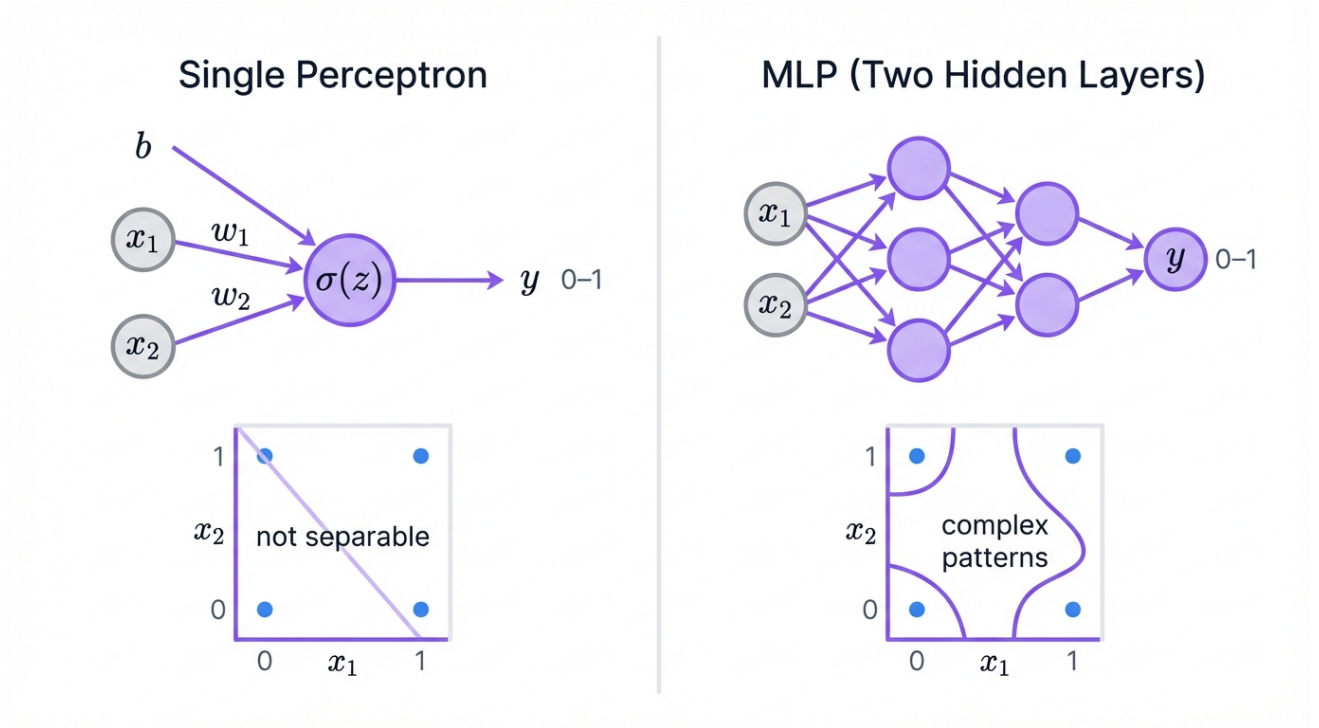


Figure 2: Left: A single perceptron, with input nodes and corresponding weights on the left and an output node on the right. Right: An MLP, which can do more complex tasks. Both have 2 input nodes on the left and a single binary classifier output node on the right.

A Convolutional Neural Network (CNN) is a type of Neural Network that uses filters or kernels to reduce the feature size of the input [31]. This type of model is often used for large modalities such as images, audio, video or 3D volumetric images. The input of the model first goes through a series of convolutional and pooling layers. Convolutional layers serve to detect and filter local features in the input. The pooling layer then merges similar features detected by the convolutional layer into one. CNNs often have two or three of these stages followed by a set of more conventional deep learning layers [25]. This structure is followed for such modalities both because it allows the model to draw on features from very different parts of the input but also because it reduces the computational complexity required to process the large input data.

A UNet is a CNN that consists of two mirrored halves and is often used for image-to-image Deep Learning tasks, as well as iterative image denoising in diffusion models [20]. One side downsamples the input image, doubling the number of feature channels at each layer; the other side upsamples to the output image, halving the number of feature channels at each layer, creating a U-shaped network [36].

The downsampling side typically consists of repeated 3x3 unpadded convolutions, each followed by a Rectified Linear Unit (ReLU) [19] and a 2x2 max-pooling layer. The upsampling side usually consists of an upsampling operation on the feature map, followed by a 2x2 convolution, halving the number of feature channels, a concatenation with the corresponding cropped feature map and two 3x3 convolutions, each followed by a Rectified Linear Unit (ReLU) activation function [36]. This

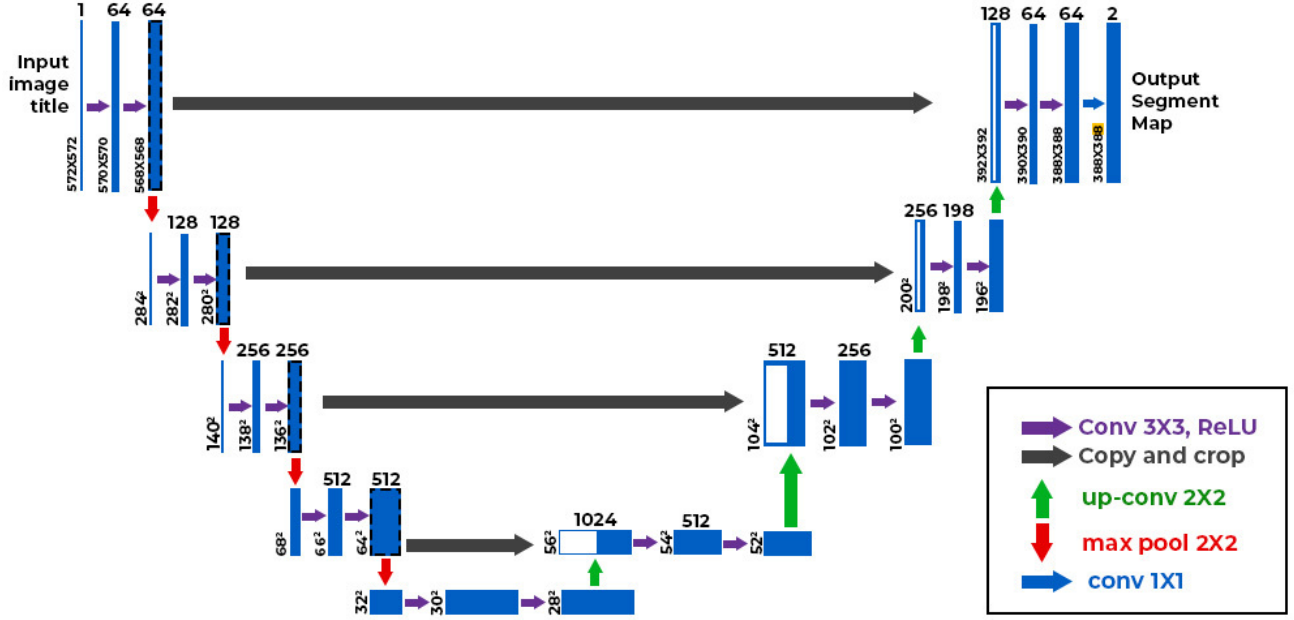


Figure 3: The UNet architecture, as described by Ronneberger *et al.* (2015) [36], with an input/output size of 64x64 and up to 1024 feature channels and skip connections at each stage.

is only a rough outline of UNets, as illustrated in Figure 3, and the exact configuration can vary widely.

Supervised training of Neural Networks usually involves feeding the input data into the model through the input layer, allowing the model to create an output and then judging that output using a loss function, which compares the ground truth "label" data to the output of the model and assigns a score for how wrong the model is. This score is then used by an optimiser to generate a set of gradient values, which can then be used to update the weights and biases of the model to move them towards an optimum, which should give a better output. This is done for the whole dataset and many times over until the model is capable of creating an accurate output [2].

2.4 Hyperparameters and Bayesian Optimisation

Neural networks are functionally a "black-box" in that, from the outside, it's difficult to understand what the model is doing to arrive at a particular output [12]. This is why optimising neural networks is considered a black-box problem: a set of parameters needs to be adjusted so that the output moves towards the desired goal, without necessarily knowing how the parameters affect the output. The model's parameters, such as the weights and biases of each node, are usually adjusted by an optimiser, such as Adam, using backpropagation [3]. While it might be considered that this solves the problem of parameter optimisation, it also creates a new problem, in that these techniques have a lot of their own parameters, usually referred to as "hyperparameters", i.e. parameters about parameters, which also need to be optimised. These include the "Learning Rate", the rate at which the parameters are adjusted towards an optimum, the number and width of layers in the model and the types of activation function and optimiser used in the training process.

The challenge then is to find a suitable method by which sets of hyperparameters can be

tested, evaluated and new ones chosen efficiently, to minimise the time and resources spent during optimisation [14]. One such method, which seems suitable for this particular application, is Bayesian Optimisation [41]. This method uses a "Bayesian Prior" such as a Gaussian function to model the objective function; an acquisition function then calculates which point in the objective space will return the most information about the objective function. These are then tested by calling the objective function, in this case the model's training function, before the prior is updated with the resulting information, creating a posterior probability. The first guess will probably be quite bad, but in relatively few steps, a relatively efficient optimum can be found [15].

3 Related Work

While the forward problem, as described in section 2.2, might be straightforward to solve using an analytical approach, the inverse problem is significantly less so. Previously, current reconstruction was performed using purely mathematical methods, such as Fourier analysis and Bayesian Inference [29]. These methods can provide a fair approximation of the current density, but aren't particularly robust to noise and can't handle the convolution introduced by SOT or larger standoff distances between the sample and the sensor, resulting in blurry and inaccurate images.

MAGIC UNet was the first attempt to solve this problem using machine learning and made significant improvements over previous methods [33]. This paper was specifically for images taken using NV imaging, which differs significantly from SOT; therefore, this thesis aims to adapt this framework to SOT.

3.1 Current Reconstruction by Fourier Analysis

One of the simplest methods for solving the inverse problem, first described by Roth *et al.* (1984) [37], is the Fourier Transform. This is a straightforward and widely used approach to current density reconstruction based on the Fourier Series, which gives a fair approximation of current density from the magnetic field. The relationship between the current density $J(r)$ (lying in a thin sheet of thickness d), and the magnetic field $B(r)$ in the x , y or z planes holds as:

$$J_y(k_x, k_y) = -\frac{2}{\mu_0 d} \frac{k_y}{k_x} e^{\sqrt{k_x^2 + k_y^2} z} b_x(k_x, k_y, z), \quad (4)$$

where k is the spatial frequency.

3.2 Bayesian Inference Method

The Bayesian inference method, as described by Clement *et al.* (2021) [11], aims to learn the best-fit current density by maximising the posterior probability $p(J|\phi)$, where ϕ is the magnetic field image. Here ϕ could be an individual component $\phi = B_z$ for images from out-of-plane methods or $\phi = \{B_x, B_y, B_z\}$ for vector magnetometry. This method can be formulated as an optimisation problem to recover the underlying current-density signal and suppress noise. It is possible to use different priors, such as a Gaussian distribution, Total Variance or Finite Support, depending on which prior assumption best supports the application [11].

This method has proven to be superior to the Fourier method, especially on out-of-plane field data, due to Bayesian Inference being able to better reject noise and provide higher-resolution reconstruction [29]. It does require the user to choose an appropriate prior for the given input.

3.3 MAGIC UNET

This thesis draws from and builds on the work of Reed *et al.* (2025)[33], who created the MAGnetic Inverse Calculation UNet (MAGIC UNet), which can predict a current density image based on a magnetic field image obtained using NV imaging and creates demonstrably better results than the previous Fourier Method. To achieve this, synthetic data was generated using COMSOL Multiphysics and then used to train a UNet.

3.3.1 Normalisation

The synthetic data used to train the MAGIC UNet models includes a large range of pixel values, which are sometimes many orders of magnitude apart, making them unsuitable to train on directly. Images with larger values may perform well on the trained network, whereas images with smaller values likely won't. In addition, larger pixel values can cause the loss values and subsequent gradients to explode, further complicating learning. To ensure that all images are learned equally well and that the model learns well overall, it is necessary to normalise individual pixel values.

This can be done by using a scalar value for each image i , a_i . The corresponding label image is then also scaled by a_i to maintain the relationship between the magnetic field density amplitude in the input image and the current density amplitude in the label image. A global scaling factor β is also applied to the label data as shown in Eq. 5, and the inverse $1/\beta$ is applied to the output of the model. In the paper by Reed *et al.* (2025)[33], β was set at $\sim 10^8$, but this value can vary.

$$Norm(J) = \frac{J}{\max(\sqrt{J_x^2 + J_y^2 + J_z^2})} \times \beta \quad (5)$$

3.3.2 Network Architecture

UNet largely follows the UNet architecture as set out by Ronneberger *et al.* (2015)[36]. The skip connections seen in Figure 4(b) are used both to preserve the image structure and to help resolve the vanishing gradients problem by allowing gradients to flow from the downsampling layers to the upsampling layers.

3.3.3 Training

The training process uses a simple Mean Squared Error (MSE) loss function, as defined in Eq. 6, which allows the network to see how wrong the prediction was compared to the label. This loss value is then used to create gradients which adjust the weights and biases of the model.

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2, \quad (6)$$

where Y is the prediction, $\hat{Y}$ is the label and n is the number of data points [43]. The loss value is then used to do backpropagation through the network before an Adam optimiser is used on the model.

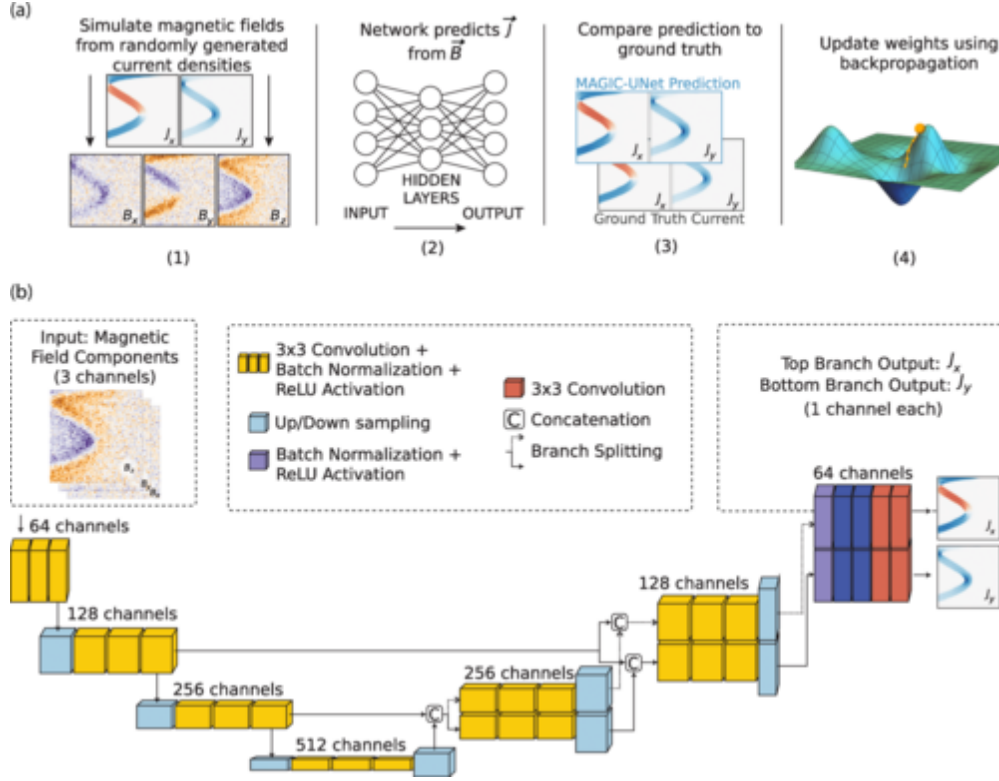


Figure 4: **MAGIC UNET** [33]. (a) Shows the data generation and model training process of MAGIC UNet. (b) Shows a diagram of the UNet architecture, adapted to this task.

4 Methods

This thesis also uses a UNet with parameters similar to those of MAGIC UNet but adapts the architecture for SOT images. The data generated for MAGIC UNet is adapted using 2 different methods to mimic SQUID data, before it's normalised and noise is added.

4.1 Data

The training data used in this thesis all came from MAGIC UNet and was adapted to enable the models to work on SQUID images rather than NV. This data was generated by Reed *et al.*[33] using COMSOL Multiphysics and MATLAB to simulate the current density in a given material, as shown in Fig. 5(a), along with the corresponding magnetic field via the Biot-Savart law, as shown in a magnitude image in Fig. 5(b) and in individual components in Fig. 5(e). The simulated conductive material here includes a variety of different wires, closely spaced layers, and non-uniform current density around corners, which should ensure that the model is trained to become fairly robust.

During training, not all data could be held in memory simultaneously, so the images were stored in files of 1024 each. These files were loaded into memory one by one and converted either to the angle at which the SQUID measures the magnetic field, through a simple angle conversion, as seen in Fig. 5(c), or through the `squidtools` package, into a more realistic SQUID image, as shown in Fig 5(d). Once the image has been converted, the data is then normalised so that the values of all the images are in roughly the same range and the model can use them without creating exploding

or vanishing gradients and making it difficult for the model to learn.

4.1.1 SQUID Angle conversion

The SQUIDs that were used to create the images in this thesis measured the field at an angle of 62° , so to make sure that the model learned the relationship between the magnetic field as seen at this angle and the underlying current density, the training data, which is split into x , y and z components, must be converted to this angle. This was done by taking only the x and z components of the data and multiplying them by the sin and cos of 62 and adding the products together: $\sin(62) \cdot B_x + \cos(62) \cdot B_z$. This then creates a single component image at the angle of the SQUID.

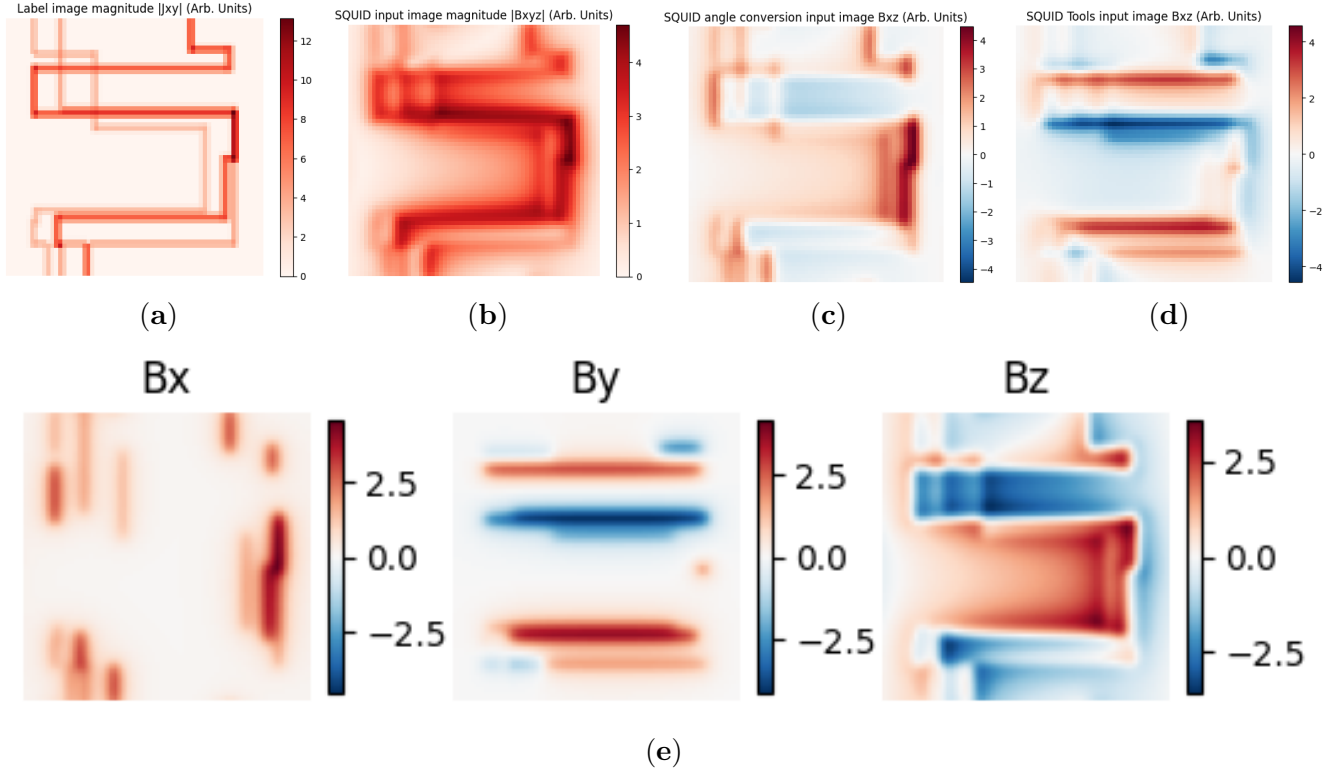


Figure 5: (a) Shows the current density image, which is used as a label during training. (b) Shows a magnitude image of the individual components of the image, which are shown separately in (e). (c) Shows a simple angle conversion of the x and z components. (d) Shows a `squidtools` conversion of the x and z components.

4.1.2 SQUID Tools

The `squidtools` Python package, created by Matthijs Rog (2026) [34], can create more realistic SQUID images than through only Biot-Savart and angle conversion, as explained in Sec. 2.2. Rather than simply taking the magnetic field input image from MAGIC UNet and converting the angle, here the label current density image was taken and fed into the `squidtools` package, along with a set of parameters for the SQUID, such as height above the sample and the inner and outer

diameter. The package then created a magnetic field image which could be used as an input image for the model, allowing it to learn a better approximation for the relationship between SQUID image and current density.

4.1.3 Pre-processing

Once the data has been converted into the appropriate format for the model, it is first normalised using the simple formula $\hat{x} = (x - \mu)/\sigma$, where x is the original feature, $\hat{x}$ is the normalised feature and μ and σ are the mean and standard deviation of the feature, respectively. Once the feature is normalised, Gaussian noise is added to the data to make sure that the model trains to be resistant to such noise.

4.2 Model Architecture

This thesis largely followed the architecture set out in the MAGIC UNet paper with a few adjustments to allow for Hyperparameter Optimisation (HPO), including an adjustable layer depth, ranging from 3 layers to a maximum of 7 layers, as well as batch size, learning rate and the number of epochs. MAGIC UNet also used Average Pooling layers rather than Max Pooling layers, so this option was added to the architecture to see if it made a difference in how well the model learns.

Replicating the results from MAGIC UNet using their code turned out to be quite challenging, so instead, the `torch` Python package was used to recreate the same architecture. This was both faster and easier to use than the `tensorflow` code that MAGIC UNet used.

4.3 Training

Before training begins, the validation and testing datasets are collected from all of the data files; these are 10% of the dataset each and can be used after each training run to assess how well the model performs.

The training process takes the magnetic field input images, which have been converted to SQUID images, normalised, and had noise added during preprocessing, and feeds them into the network, before looking at the output of the model and comparing it to the label current density image. This comparison is done using a loss function, in this case, MSE, which creates a loss value, allowing the network to create gradients which adjust the weights and biases of the model during backpropagation. Finally, the Adam optimiser is used to optimise the model during the backward pass. This process is repeated for the whole training set and for as many epochs until the loss and accuracy values plateau.

4.4 Hyperparameter Optimisation

As mentioned, machine learning models have a set of hyperparameters that need to be tuned. Rather than trying to find the optimal hyperparameters for the architecture and dataset manually, a Hyperparameter Optimisation (HPO) technique is applied; in this case, Bayesian Optimisation.

The `gp_minimize` function was used from the `scikit-optimize` Python package, which applies a Gaussian Process minimisation function [15] to a defined search space with an objective function.

The search space in this instance was defined with 5 hyperparameters: the number of layers, the learning rate, the batch size, the number of epochs and the noise level.

To ensure that the BO function doesn't just choose the lowest noise level to minimise the loss value, the evaluation function measures the loss at different noise levels from $\sigma = 1e - 4$ to $\sigma = 1$. Then it averages the loss to return a fairer representation of what the model is capable of.

4.5 Evaluation

To evaluate the capabilities of the models, 3 main measures are used: Root Mean Squared Error (RMSE), Structural Similarity Index Measure (SSIM), and Peak Signal-to-Noise Ratio (PSNR).

4.5.1 RMSE

Root Mean Squared Error is simply the square root of the MSE loss function that we use to train the model, as shown in 7. As a loss function, the goal is to minimise this measure [22].

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \quad (7)$$

4.5.2 SSIM

The Structural Similarity Index Measure is a metric for measuring the preservation of structural information. This means it is possible to quantify how well the image is translated from the SQUID image to current density. SSIM incorporates perceptual phenomena, including luminance and contrast masking terms, to consider image degradation as a perceived change in structural information, as shown in Eq. 8. This distinguishes it from MSE or PSNR, which estimate absolute errors.

$$SSIM(x, y) = \frac{(2\mu_x, \mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}, \quad (8)$$

where μ is the sample mean of x or y , σ^2 is the sample variance of x or y , σ_{xy} is the sample covariance of x and y . $c_1 = (k_1 L)^2$ & $c_2 = (k_2 L)^2$ are two variables that stabilise the division with a weak denominator; L is the dynamic range of the pixel values (often $2^n - 1$, where n is the number of bits per pixel) and finally $k = 0.01$ and $k = 0.03$ by default [42].

5 Experiments & Results

5.1 Comparison between CDIP and MAGIC UNet

This section will compare the performance of the previous MAGIC UNet paper to the model described here at various stages. This was done by taking the SSIM, RMSE and PSNR measurements, as described. These comparisons are not entirely direct, as MAGIC UNet takes 3 components as input, while CDIP takes only 1, which makes the task harder, as the model will have to find the same information with only a third of the information. Furthermore, to be able to take the measurements, the image has to be only a single component, and the output of both models is 2 components, so the magnitude of the image is calculated as $|J| = \sqrt{J_x^2 + J_y^2}$.

5.1.1 Quantitative results

Model	SSIM	RMSE
MAGIC_UNET	0.924	2749
CDIP	0.923	0.400

Table 1: **Quantitative comparison between MAGIC UNet and CDIP.** CDIP compares almost exactly with MAGIC UNet, with MAGIC UNet being perhaps slightly ahead on SSIM.

Table 1 shows the results of the quantitative analysis of both models, in which they were assessed using the "Out-of-distribution" dataset from the MAGIC UNet paper using both SSIM and RMSE. The RMSE figures for MAGIC UNet were very high, probably because of the normalisation method used, where the prediction is de-normalised afterwards.

5.1.2 Qualitative results

Figure 6 shows an image generated by MAGIC UNet, along with the accompanying quiver plot and streamline graphs, which show the direction in which the current is travelling. Here, the input image has a noise of $\sigma = 1$, to show how resistant the model is to noise. The scale on these images is also very large due to the normalisation method.

In Figure 7, we can see the same input and label image as in Fig. 6, but this time the prediction is created using the CDIP model, which has been trained on data that was converted using the simple angle conversion technique. CDIP performs slightly worse on this training data, with some of the directions at the edge of the image curving away from the intended direction.

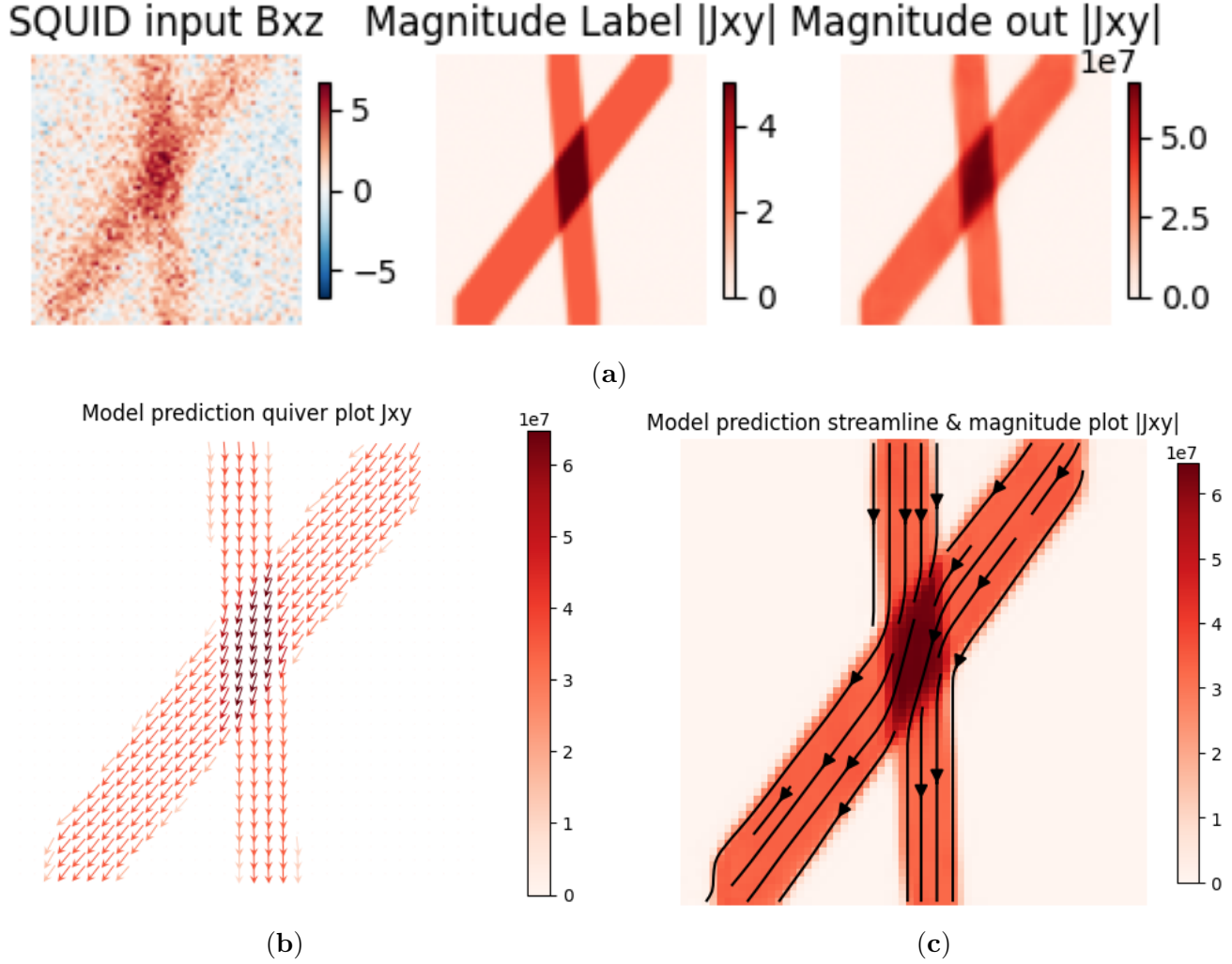


Figure 6: **Image generated by MAGIC UNet.** (a) Shows a comparison between the input image on the left, with a noise of $\sigma = 1$, the magnitude of the current density label image in the centre and the magnitude of the prediction on the right. (b) Shows a Quiver plot of the predicted image and (c) shows a streamline plot of the predicted image.

5.2 Comparison between CDIP and Fourier Analysis

This experiment repeats the analysis that MAGIC UNet did in comparing the Deep Learning techniques used to the performance of Fourier Analysis [33]. The expectation is that the same techniques will again outperform Fourier Analysis due to predictions of SOT data being a more difficult task than the NV data used in MAGIC UNet and the fact that MAGIC UNet already comfortably outperforms Fourier Analysis [33].

5.2.1 Quantitative results

Table 2 shows the results between Fourier Analysis and CDIP, trained with Angle Conversion data. It is very clear, looking at the SSIM scores, that while CDIP creates an image that is visually

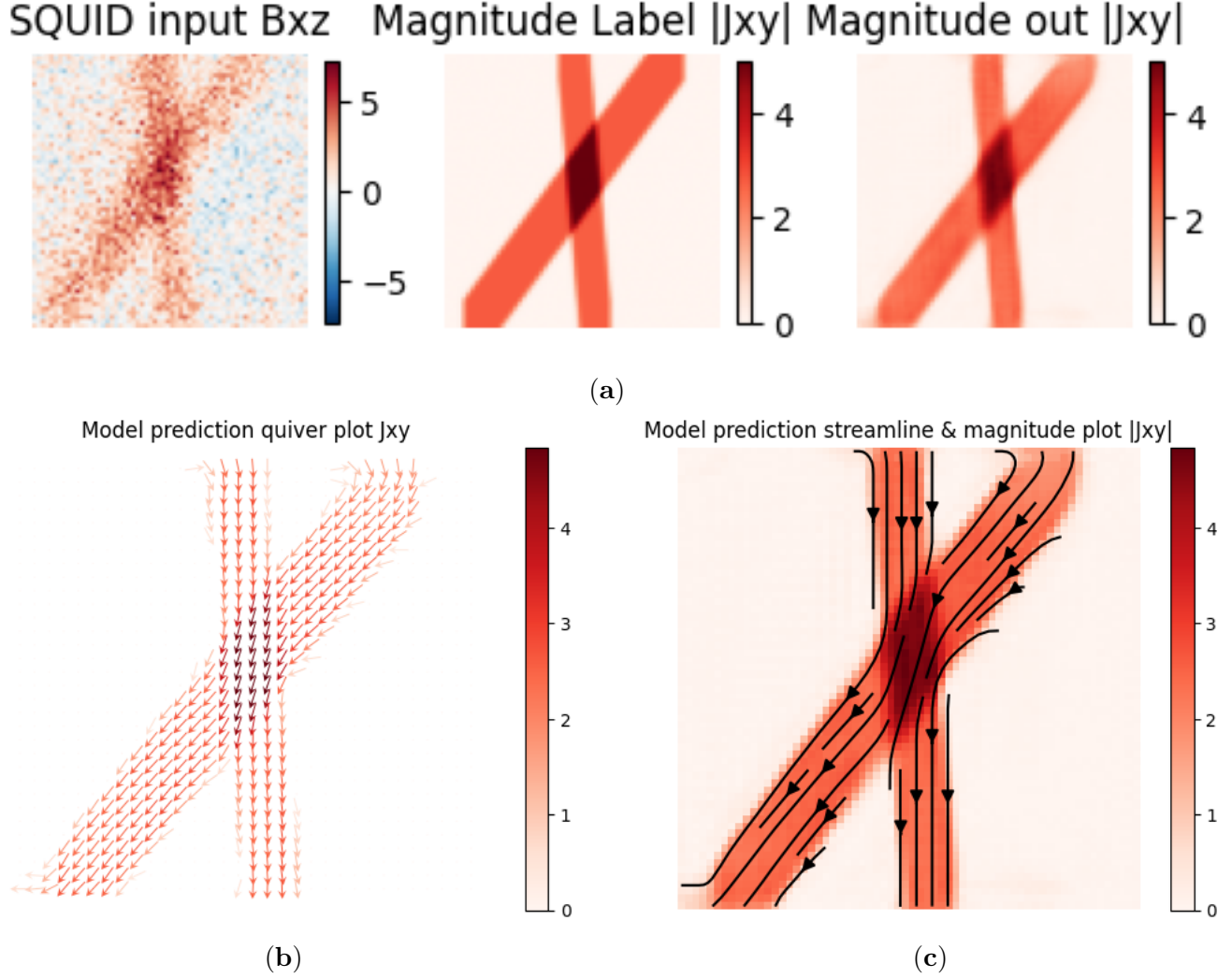


Figure 7: **Image generated by CDIP trained on angle conversion data.** (a) Shows a comparison between the input image on the left, with a noise of $\sigma = 1$, the magnitude of the current density label image in the centre and the magnitude of the prediction on the right. (b) Shows a Quiver plot of the predicted image, and (c) shows a streamline plot of the predicted image.

similar to the ground truth, the same can't be said for Fourier Analysis. The large RMSE also reflects the fact that despite the input values being normalised beforehand, Fourier returns an image with very high values.

Model	SSIM	RMSE
Fourier	0.031	116979423099.362
CDIP Angle	0.923	0.400

Table 2: **Quantitative comparison between Fourier Analysis and CDIP.** CDIP gets a much higher SSIM score and a more reasonable RMSE than Fourier.

5.2.2 Qualitative results

Figure 8 shows the comparison between the prediction of CDIP and Fourier Analysis on the same input image, which was converted using angle conversion and has $\sigma = 1$ noise added to it. It's clear by looking from comparing the CDIP and Fourier images to the Label image in Fig. 8(b) that the CDIP prediction is much closer to the ground truth.

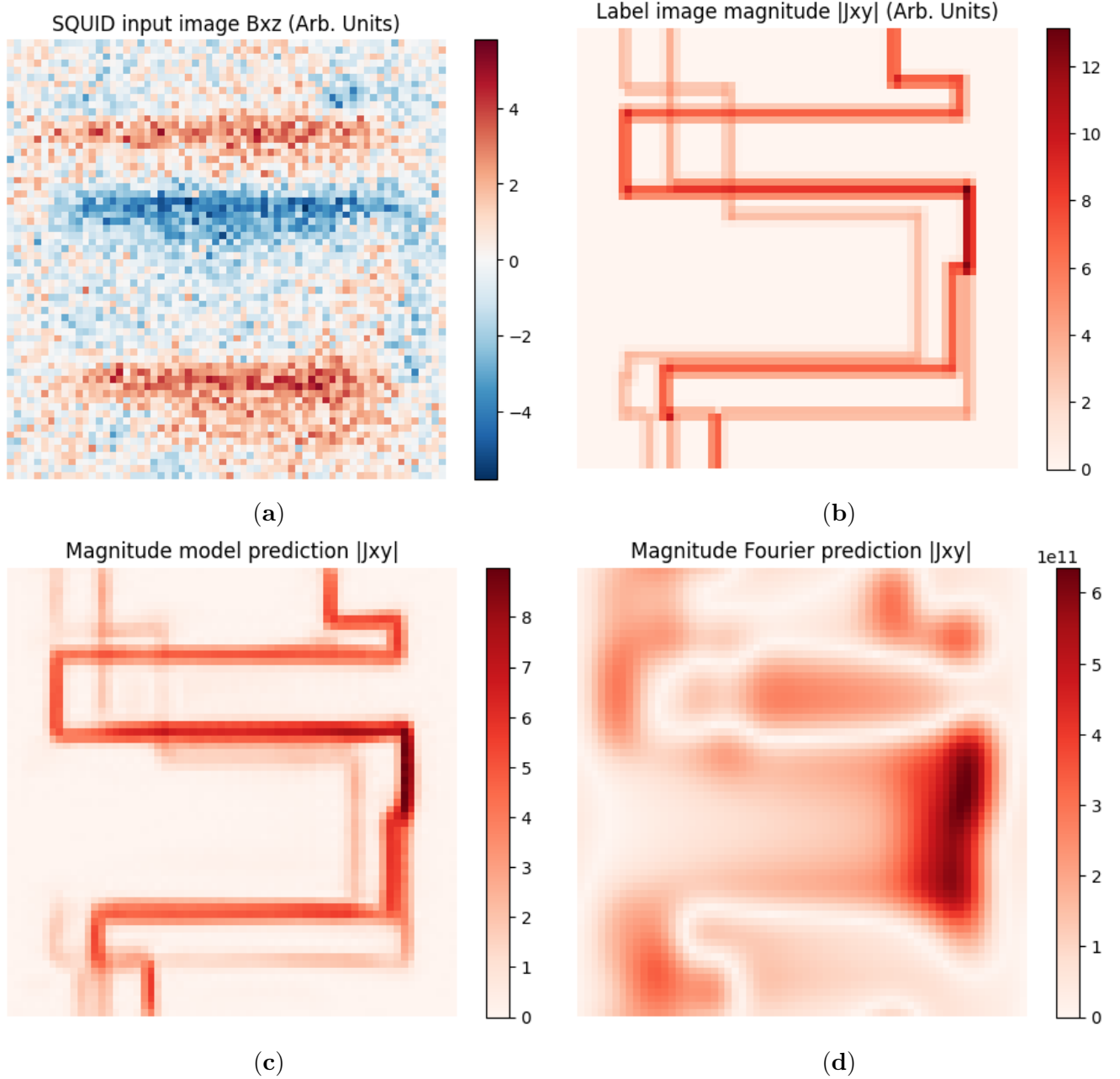


Figure 8: **Images generated by Fourier Analysis and CDIP trained on angle conversion data.** (a) Shows the input image with the simple angle conversion and $\sigma = 1$ noise. (b) Shows the magnitude image of the label current density. (c) Shows the prediction made by CDIP, trained on angle conversion data. (d) Shows the prediction made using Fourier Analysis.

5.3 Comparison between squidtools and angle conversion

This experiment shows the difference between and compares the performance of CDIP trained using angle conversion and squidtools data. In theory, squidtools data should more closely reflect the real images that a SQUID creates.

5.3.1 Quantitative results

Table 3 shows the comparison between CDIP models trained using Angle Conversion and Squidtools data. Both models do relatively well on both SSIM and RMSE measures, with the models being relatively closely matched. On both measures, it is clear, however, that the Squidtools model does slightly worse than the Angle Conversion model.

Model	SSIM	RMSE
CDIP Angle	0.923	0.400
CDIP squidtools	0.916	0.417

Table 3: **Quantitative comparison between CDIP trained on Angle conversion and Squidtools.** Angle Conversion and SquidTools have similar SSIM and RMSE scores, but Angle Conversion edges ahead on both measures.

5.3.2 Qualitative results

Figure 9 shows the prediction from the angle conversion model, and Figure 10 shows the prediction from the Squidtools model, both on a real SOT image, downsized to fit the model input of 64x64 and normalised. The angle conversion model can predict the current much more clearly, with less noise and much less spillover between the different wires in the serpentine, than the Squidtools model. Both models do have a little spillover and also some circular currents.

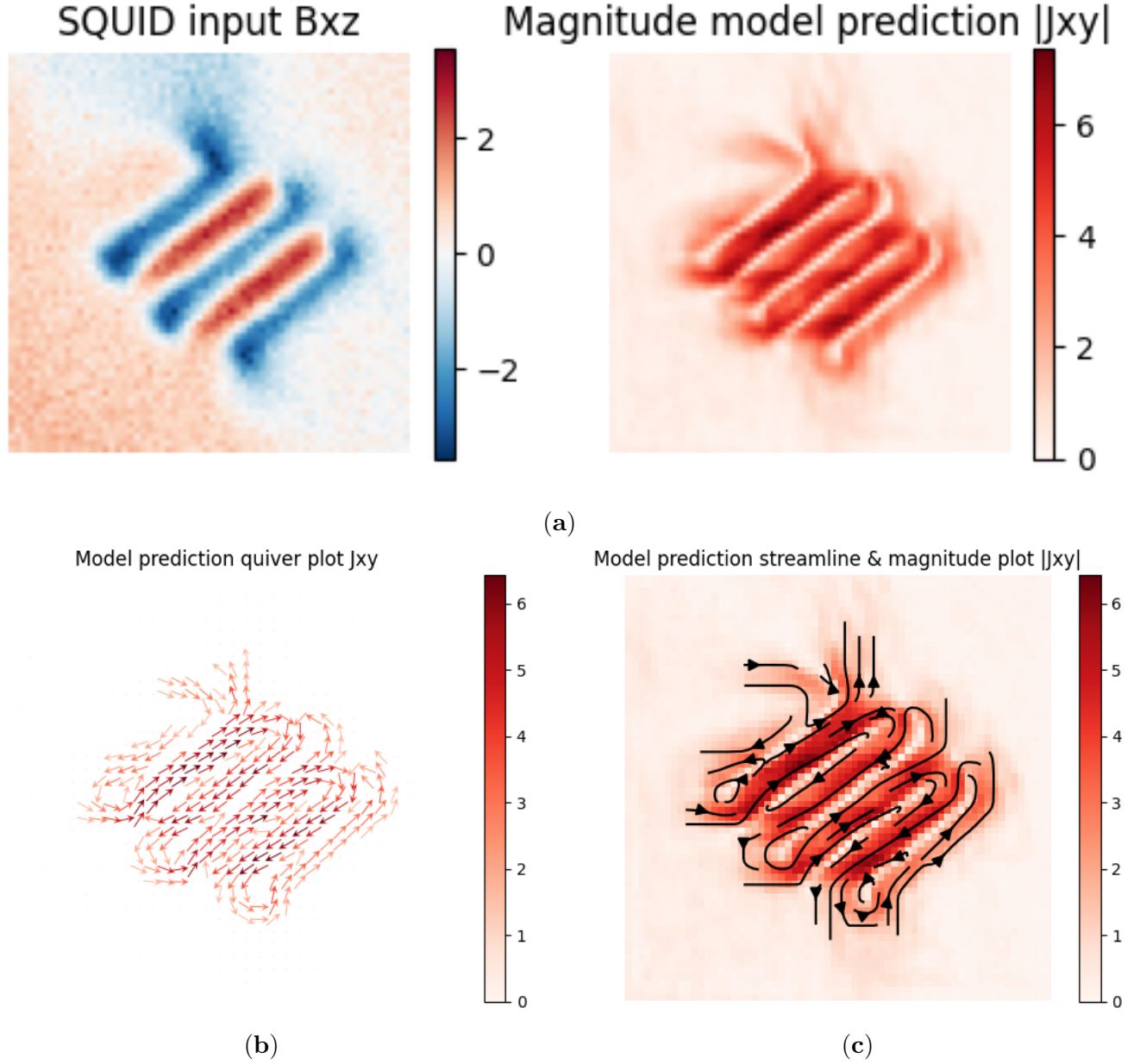


Figure 9: **Real SOT image. Current prediction by CDIP trained on angle conversion data.** (a) Shows a comparison between the input image on the left, with a noise of $\sigma = 1$, the magnitude of the current density label image in the centre and the magnitude of the prediction on the right. (b) Shows a Quiver plot of the predicted image, and (c) shows a streamline plot of the predicted image.

6 Discussion

The experimental results show clearly that the CDIP model performs comparably to, if slightly worse than, MAGIC UNet, despite the task being substantially more difficult, with a one-component input, taken at an angle, rather than the 3 components used by MAGIC UNet. This answers the

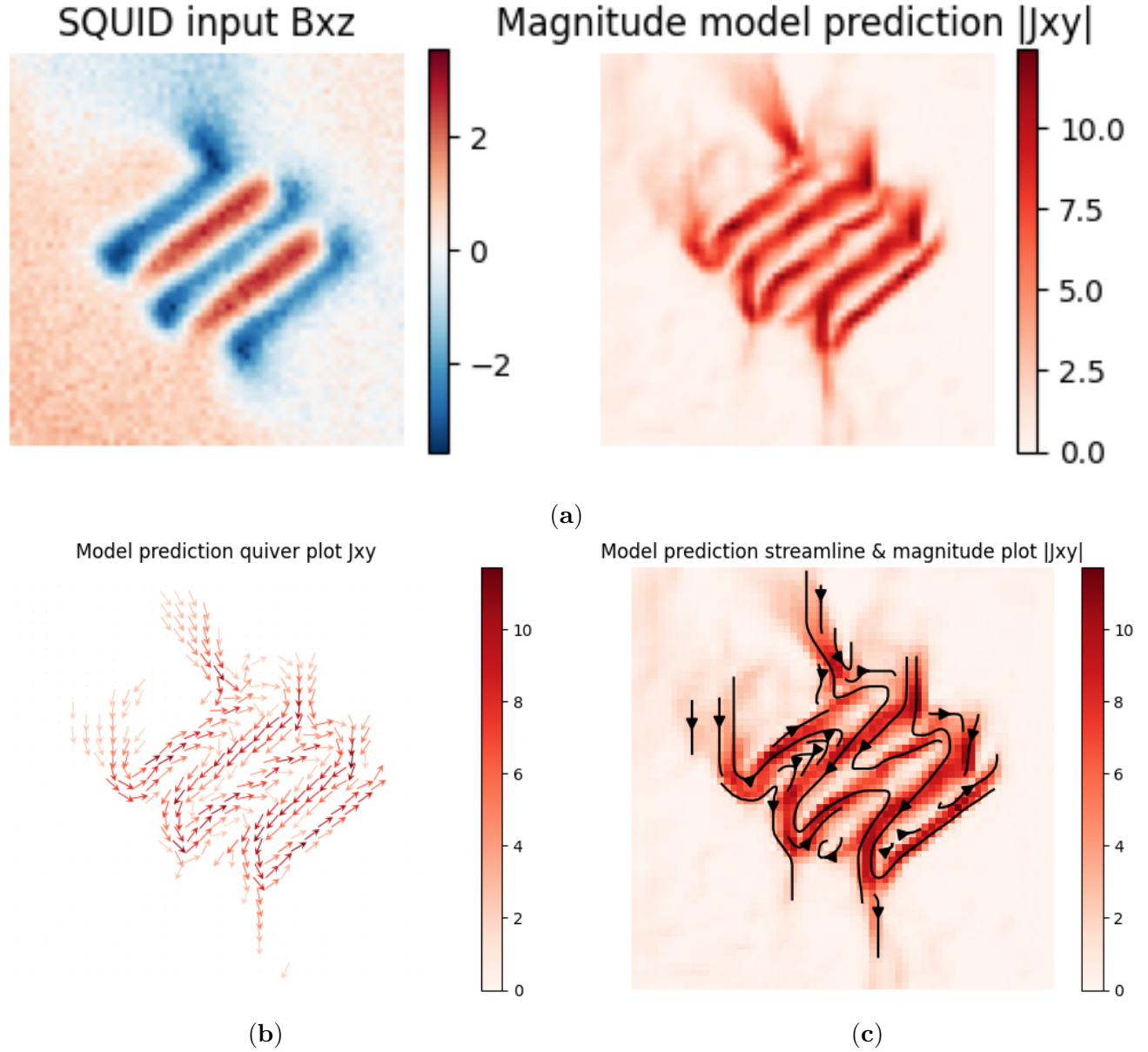


Figure 10: **Real SOT image. Current prediction by CDIP trained on squidtools data.** (a) Shows a comparison between the input image on the left, with a noise of $\sigma = 1$, the magnitude of the current density label image in the centre and the magnitude of the prediction on the right. (b) Shows a Quiver plot of the predicted image, and (c) shows a streamline plot of the predicted image.

first research question, as the UNet architecture could effectively be adapted to SOT data.

The second experiment showed that CDIP far outperformed the previously used Fourier Analysis method [37], as this method had difficulty rejecting much noise in the image, as well as having difficulty with large pixel-to-pixel gradients and uniform distributions [33]. It was also the goal of this thesis to make a direct comparison to the Bayesian Inference method; however, due to a lack of code availability and time constraints, this was not possible. Looking at the previous comparison

made to the Fourier method by Midha *et al.* (2024)[29], CDIP might be comparable to Bayesian Inference in performance, with the caveat that Bayesian Inference also relies on the user to select an appropriate prior distribution for the input data; otherwise, the result is significantly worse [11].

Finally, the last experiment showed that the CDIP model that used the simple angle conversion on the training data seemed to perform better, both on the training data and the real SOT images. It was relatively predictable that this would be the case on training data, as the task is slightly more difficult for the SquidTools model because the SQUID practically acts as a convolution on the data when compared to angle conversion. It might be expected, however, that the SquidTools model might perform better than angle conversion on real images, as it provides a more accurate training image when compared to real SOT images. This discrepancy is most likely due to the output of the SQUID, and by extension SquidTools, being very sensitive to the parameters used to create the images, especially the height and angle of the SQUID relative to the sample. It might therefore be useful to do further experimentation on this in the future.

6.1 Contribution

This thesis showed that deep learning is an effective method of current density reconstruction for SOT data, that it can outperform previously used methods, and that the method of data generation can greatly affect the performance of the model. This is a foundation for further work, which could enhance the performance of this technology and prepare it for practical and commercial use in SOT inspection.

6.2 Limitations

The main limitation of this thesis is probably the fact that no direct comparison could be made to Bayesian Inference, which makes it difficult to know how much more effective CDIP is at reconstructing current density, if at all.

Another limitation would be the aforementioned sensitivity of SquidTools and the model trained using it to changes in the parameters, as well as the difficulty of ascertaining what the parameters of the SQUID and any particular measurement are with any real precision. This means that either a lot of work needs to be done to estimate the parameters for each real SOT image and a new model has to be trained for each parameter set, or it might be possible to train a model to be resistant to changes in the SquidTools parameters by varying them during training.

Hyperparameter Optimisation was performed several times during the development of CDIP, but it was found that this did not add much in the way of extra performance, so the experiments were done with a standard set of hyperparameters, which seemed to work well, for the sake of time.

Finally, the last limitation is that, while CDIP does perform relatively well on real SOT images, it's clear that the result is not yet adequate for commercial purposes and still requires much improvement. It is also clear that there is still much work to do before these models can be trusted to accurately predict the current density in complex and unpredictable superconducting materials.

6.3 Future Work

In order to create models that can more accurately reflect the real conditions inside the material, there are several possible improvements which may make the diagnostics of quantum hardware and

superconducting materials easier.

The easiest and most obvious improvement would be to increase the resolution of the image; most of the real SQUID images in this thesis were originally 200x200 pixels, so a model of that size or even 128x128 would at least double the resolution and allow for a better understanding of the underlying current density phenomena.

Another quite simple way to improve the model might be to increase the size and variety in the dataset, to allow the model to better learn how the current behaves in more complex samples and how that is reflected in the magnetic flux image.

The previous Fourier and Bayesian methods that were used to approximate the current density, while generally seeming to be worse, might give the model a good starting point for figuring out the current density, either by giving the output of either or both methods as an input for the model or by incorporating the methods as an extra layer in the model itself.

SOT also measures the topography and temperature of the sample at the same time as the magnetic flux, so giving the model this data at the same time would make it much easier for it to learn how the current density distribution should look. Certainly it should end any stray current predictions that run outside of the material.

It has also been mentioned that it may be most straightforward to restrict the output of the model in such a way that it is constrained by the physical restrictions on the current density to force it to learn the physics and ensure that the output is at least physically possible.

There are a few different deep learning architectures that could perform this kind of prediction [24][17][9]; UNET is only the most obvious, tried and tested one. It would be interesting to see how well other architectures work on this kind of task.

In the current method of deep learning, there is no limitation to what the model can predict, meaning that the model can predict current flows, which don't necessarily follow the laws of physics. It is possible and might be advantageous to limit the output of the model to known physical boundaries or to increase the loss more when the output doesn't follow physics, so that the model better learns how physics works.

Finally, it should be possible to measure the current density at different heights [35], and therefore to predict where the current runs in 3-dimensional space. Creating a model that can make 3-dimensional predictions of the current density might be the ultimate reflection of reality and therefore a very useful diagnostic tool.

7 Conclusion

This thesis looked at the viability of deep learning techniques in Current Density reconstruction for SQUID-on-tip magnetic field data and found that these techniques can effectively be applied in this way, with results comparable to previous works even when performing a more difficult task. It also found that they are often comparable to or even outperform analytical methods, such as Fourier Analysis. A comparison was also done between two methods of converting training data into SOT-like data and found that both methods work; the SquidTools method requires either very precise parameters or more work to make the model resistant to parameter changes. While current models can predict current density in real SOT images, these are still fairly inaccurate, and there is much room for improvement in this area.

8 Reflection

Over the course of this thesis, I have had to learn a lot about the physics side of this thesis, which, as a computer scientist, wasn't always the easiest to get my head around. The biggest challenge for me was to understand what information was needed from a physics standpoint and why certain aspects were important. This thesis began over a year ago and took much longer than expected and achieved less than I had wanted. Much time was spent at the beginning trying to replicate what previous papers had shown, using their code, when I should perhaps have created my own code and saved much time. I may also have avoided a lot of stress and made my supervisors happier by looking earlier at how I presented my results and asking more about what they were looking for.

8.1 AI use

The Grammarly spelling and grammar-checking plugin was used in the course of writing this thesis. I found that this tool is often useful in correcting mistakes and improving sentence structure, although occasionally the suggestions differ from my own writing style or don't make sense.

The Claude and Gemini chatbots were occasionally used to write or debug code, usually as a last resort. The vast majority of the programming work was done manually out of preference.

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I would like to dedicate my thesis to my grandparents, who have been an immense help over the last two years. I would also like to thank my mother and siblings for their support, and especially my study coach Alexandra Blank, without whom I would have struggled much more.

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A Appendix

A.1 Technologies benefitting from CDR

Other technologies that may benefit from improved current density reconstruction include, but are not limited to: integrated circuits [30], microelectronics [1], batteries [21], solar panels [39], superconductors [10], mechanical joints [5], fault analysis [45], quantum computing [27], exotic materials [13], new semiconductors [4], and eddy current imaging [44].